

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – I
PAPER –1
TEST DATE: 30-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A

Sol. Rate of loss of water $= \int_0^2 \pi r^2 (dy \eta \sqrt{2gy})$

2. C

Sol. From the equation of trajectory $y = 10 \tan \theta - \frac{20}{9} \sec^2 \theta$

$$\frac{dy}{d\theta} = 0 \text{ gives, } \tan \theta = \left(\frac{9}{4}\right)$$

3. B

Sol. $4 \times 2\pi = \frac{1}{2} \times \alpha \times 1^2, \quad \alpha = 16\pi$

$$\text{Also, } F = \frac{mr\alpha}{2}$$

4. A

Sol. Applying conservation of angular momentum about Q

$$mvR \sin 30 = (mvd/\sqrt{3})$$

$$\therefore d = \sqrt{3}/2R$$

5. ABC

6. AD

7. ACD

- 8. AD
- 9. ACD
- 10. ACD

Section – B

11. 3

12. 2

13. 2

Sol. $t_1 = RC$; $t_2 = R/L$

$$\rightarrow LC = t_1 t_2 = 0.1 \text{ sec} \rightarrow T = 2\pi \sqrt{\frac{L}{C}}$$

$$T = 2 \text{ sec.}$$

Section – C

14. 6.67

15. 66.67

16. 0.60

17. 4.47

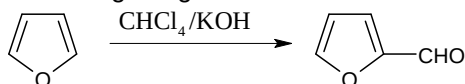
18. 21.33

19. 12.00

Chemistry**PART – II****Section – A**

20. A
Sol. Solution as progressive hydrogenation becomes easier.

21. B
Sol. Remaining will give aromatic 6 membered substance



22. D
23. C
24. ACD
25. ABD
Sol. On hydrolysis (A), (B) and (D) all give equilibrium mixture of α -D-glucose, β -D-glucose and open chain D-glucose with sp, rotation $+52.7^\circ$ where as sucrose give equimolar mixture of D-glucose and D-fructose with sp, rotation -20° .
26. BCD
27. ACD
28. ABCD
29. ACD

Section – B

30. 4563
Sol. Number of homopolymer = 4 Polythene Neopren, Teflon, PVC
Number of co-polymer = 5 Bakelite, nylon66, Buna-S, Buna-N, Dacron
Number of addition polymer = 6 Polythene, PVC, Buna-S, Buna-N, Teflon, Neoprene
Number of condensation polymer = 3 nylon66, Bakelite Dacron

31. 5
Sol. Find basic 4th 6th 8th and 9th

32. 771
Sol. $m = \frac{3/60}{250} \times 1000 = 0.2 \text{ molal}$

Therefore, K_f of water from urea solution is $K_f = \frac{\Delta T_f}{m} = \frac{273 - 272.63}{0.2} = \frac{0.37}{0.2} = 1.85 \text{ K molal}^{-1}$

Now, we determine the molality of blood by using:

$$\Delta T_f = K_f m \Rightarrow 273 - 272.44 = 1.85 \times m \Rightarrow m = \frac{0.56}{1.85} = 0.303 \text{ molal}$$

Colligative properties are defined for dilute solutions by assuming molarity = molality.

Therefore, molality = molarity = 0.303. Now using $\pi = CRT$, we get

$$\pi = 0.303 \times 0.0821 \times 310 = 7.7116 \text{ atm}$$

Section – C

33. 0.98

34. 1.00

Sol. (for Q. 33 – 34)

 Under low pressure region, V is high ($V-b \approx V$)

$$\left(P + \frac{a}{V^2}\right)V = RT$$

$$PV + \frac{a}{V} = RT$$

$$\therefore Z = \frac{PV}{RT} = \left(1 - \frac{a}{RTV}\right)$$

$$= 1 - \frac{20.39}{0.0821 \times 350 \times 35} = 0.98$$

 Under high pressure region, P is high $\left(P + \frac{a}{V^2}\right)V \approx P$

$$\therefore P(V-b) = RT \text{ or } PV - Pb = RT$$

$$\text{Or } \frac{PV}{RT} - \frac{Pb}{RT} = 1$$

$$\Rightarrow Z = \frac{PV}{RT} = 1 + \frac{Pb}{RT}$$

$$= 1 + \frac{b}{V} \text{ by approximation } \left(\because \frac{PV}{RT} = 1, \frac{P}{RT} = \frac{1}{V}\right)$$

$$= 1 + \frac{0.1383}{35} = 1 + 0.004 = 1.004$$

35. 0.11

36. 0.14

Sol. (for Q. 35 – 36)

$$\begin{aligned} \therefore K_{C_1} &= \frac{[B][C]^2}{[A]} = \frac{(a+3b)(2a-b)^2}{(1-a)} \\ &= \frac{[0.25 + (3 \times 0.167)][0.5 - 0.167]^2}{(1-0.25)} = 0.11 \end{aligned}$$

$$\begin{aligned} \therefore K_{C_2} &= \frac{[D]^2[B]^2}{[C]} = \frac{(2b)^2(a+3b)^3}{(2a-b)} \\ &= \frac{[2 \times 0.167]^2 [0.25 + (3 \times 0.167)]^3}{(0.5 - 0.167)} = 0.142 \end{aligned}$$

37. 0.21

38. 0.32

Mathematics**PART – III****Section – A**

39. A

Sol. $x + \sqrt{x^2 + 1} = \frac{1}{y + \sqrt{y^2 + 1}}$

$$\ln(x + \sqrt{x^2 + 1}) = \ln\left(\frac{1}{y + \sqrt{y^2 + 1}}\right)$$

$$x + y = \sqrt{x^2 + 1} + \sqrt{y^2 + 1} \dots \dots \dots (1)$$

Similarly, $y + \sqrt{y^2 + 1} = \frac{1}{x + \sqrt{x^2 + 1}}$

$$x + y = -\sqrt{x^2 + 1} - \sqrt{y^2 + 1} \dots \dots \dots (2) \text{ adding 1 and 2 } x + y = 0$$

Image of $xy = 1$ in $xy = 1$ itself

40. B

Sol. $\frac{1}{r_1}, \frac{1}{r_2}, \frac{1}{r_3}$ are in A.P.

$$\frac{(s-a)}{\Delta}, \frac{(s-b)}{\Delta}, \frac{(s-c)}{\Delta} \text{ are in A.P.}$$

a, b, c in A.P.

41. C

Sol. $|A|^{(n-1)^2} = |A|^{(2n^2 - 7n + 7)}$

$$n^2 - 2n + 1 = 2n^2 - 7n + 7$$

$$n^2 - 5n + 6 = 0$$

$$\Rightarrow n = 3, 2$$

42. D

Sol. Let $I = \int_0^3 \frac{f(x)dx}{(x^2 - 3x + 1)} \dots \dots \dots (1)$

$$I = \int_0^3 \frac{f(3-x)}{(x^2 - 3x + 1)} \dots \dots \dots (2)$$

$$(1) + (2) \Rightarrow 2I = 3$$

$$I = \frac{3}{2}$$

43. C

Sol. $V_n = 2a_{n+2} + S_n$

$$= 2[-4 + (n+1)] + \frac{n}{2}[-8 + (n-1)]$$

$$= \frac{n^2 - 5n - 12}{2}$$

V_n is minimum at $n=2$ or $n=3$

$\Rightarrow$ Minimum value of V_n is -9

44. ACD

Sol. $8^3x^9 + 9 \cdot 8^2x^6 + 27 \cdot 8x^3 - x + 219 = 0$

$$[8x^3 + 3]^3 = 512x - 192$$

$$x^3 + \frac{3}{8} = \left(x - \frac{3}{8}\right)^{1/3}$$

$$\Rightarrow f(x) = f^{-1}(x)$$

$$\Rightarrow f(x) = x[\cdot \cdot f(x) \text{ is continuous and } \uparrow \text{ fn}]$$

$$\Rightarrow x^3 + \frac{3}{8} = x$$

$$x = \frac{1}{2}, \frac{-1 + \sqrt{13}}{4}, \frac{-1 - \sqrt{13}}{4}$$

45. ABCD

Sol. $x^3 + 2x^2 - 3x + 1 = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

$$x^3 - 3x^2 - 2x + 1 = 0 \begin{cases} 1/\alpha \\ 1/\beta \\ 1/\gamma \end{cases}$$

Given expression $\frac{1}{3 - \frac{1}{\alpha}} + \frac{1}{3 - \frac{1}{\beta}} + \frac{1}{3 - \frac{1}{\gamma}}$

$$= \frac{\sum \left(3 - \frac{1}{\alpha}\right) \left(3 - \frac{1}{\beta}\right)}{\left(3 - \frac{1}{\alpha}\right) \left(3 - \frac{1}{\beta}\right) \left(3 - \frac{1}{\gamma}\right)} = \frac{27 - 6 \sum \left[\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right] + \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}}{\left(3 - \frac{1}{\alpha}\right) \left(3 - \frac{1}{\beta}\right) \left(3 - \frac{1}{\gamma}\right)} = \frac{11}{7}$$

46. AD

Sol. Given $f(f(x)) = x$

$$\Rightarrow \frac{af(x) + b}{cf(x) + d} = x$$

$$\Rightarrow f(x) = \frac{-dx + b}{cx - a} \dots \dots \dots (1)$$

Given $f(x) = \frac{ax+b}{cx+d}$ (2)

from(1)&(2) $a=-d$ (3)

$f(19)=19$

$\Rightarrow 38a+b=97 \times 97c$ (4)

$f(97)=97$

$\Rightarrow 194a+b=97 \times 97c$ (5)

from (5)-(4)

$a=58c$

$\therefore a=58=2^1 \cdot 29^1$

47. BD

Sol. $f(a_1) \leq f(a_2) \leq f(a_3) \leq \dots \leq f(a_{100})$ It is equivalent to distribution of 100 objects to 50 persons such that each person get atleast one object.

$\therefore$ number of such mappings $= {}^{99}C_{49} = {}^{99}C_{50}$

48. AB

Sol. $x^3 - x^2 + bx + c = 0$ (1)

Let the roots of the equation be $a-d, a, a+d$

Sum of the roots $= 1$

$a-d+a+a+d=1$ or $a=1/3$ (2)

$(a-d)a+a(a+d)+(a-d)(a+d)=b$

or $3a^2 - d^2 = b$

or $3(1/9) - d^2 = b$ or $1/3 - d^2 = b$

or $d^2 = 1/3 - b$

Since d is real, $1/3 - b \geq 0$

$b \leq 1/3$

thus $b \leq 1/3$

Product of roots $= -c$

$(a-d)a(a+d) = -c$

$a(a^2 - d^2) = -c$

$1/3(a/9 - d^2) = -c$

as d is real

$d^2 \geq 0$

$\Rightarrow c \geq -\frac{1}{27} \Rightarrow -\frac{1}{27} \leq c < \infty$

Section – B

49. 8

Sol. The triangle has circumcentre at origin and its orthocentre laying on the circumcircle.

50. 5

$$\begin{aligned} \text{Sol. } \lim_{y \rightarrow 1} \frac{100y^{100}}{y^{100}-1} - \frac{50 \cdot y^{50}}{y^{50}-1} \\ \lim_{y \rightarrow 1} 100 - 50 - \left(\frac{100}{1-y^{100}} - \frac{50}{1-y^{50}} \right) \\ L = 50 - L \therefore L = \frac{50}{2} = 25 \end{aligned}$$

51. 3

Sol. No condition on l, j, k

$$\frac{3}{2} \times \frac{3}{2} \times \frac{3}{2} = \frac{27}{8}$$

When any two of l, j, k are equal

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^{2i}} \cdot \frac{1}{3^j} = \frac{3}{2} \left[1 + \frac{1}{9} + \dots \right]$$

$$\frac{3}{2} \times \frac{9}{8} = \frac{27}{16}$$

 Where all three equal $\frac{27}{16}$

$$\text{Required } \frac{27}{8} - 3 \cdot \frac{27}{16} + \frac{2 \cdot 27}{26} = \frac{81}{208}$$

Section – C

52. 2.00

53. 4.00

54. 0.00

55. 4.00

56. 1.11

57. 0.00